

RE-RUN: THE CAMPUS RUNNER

Official Game Build & Technical Documentation | Godot 2D Engine v4.3

1. Executive Summary & Build Overview

RE-RUN is a high-octane 2D pixel-art endless runner game developed in the **Godot Engine (v4.3)**. The player assumes the role of a college student sprinting through dynamic campus environments, dodging traffic barriers and overhead pipes, hitting *Mario-style Question Blocks* to solve academic quizzes, and battling departmental Professor Bosses to advance through university levels.

GitHub Repository:	https://github.com/stewin16/RE-RUN
Engine Target:	Godot Engine 4.3 (Windows x64 Portable)
Launcher Script:	PLAY_GAME.bat (One-click launch)
Asset Packages:	KW_School_Character_Set_Free.zip, pixel_art_clouds.zip

2. Game Visuals & Interface Screenshots

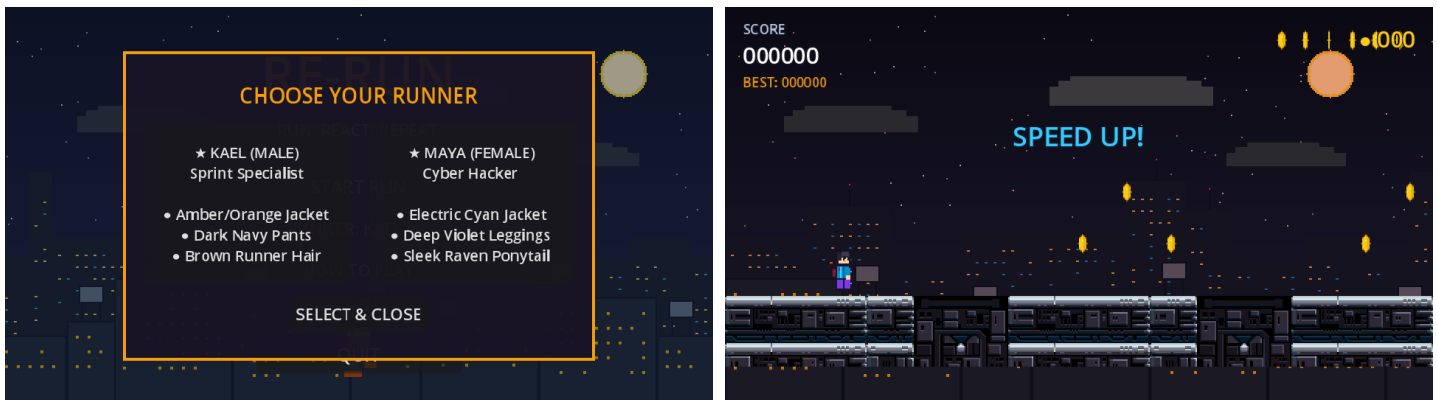


Figure 1: Main Menu Character Selection (Left) & Active Gameplay with Parallax Sky (Right)

3. Core Gameplay & Keybinding Controls

Action / Function	Keybinding / Controls	Mechanic Description
Run / Horizontal Move	A / D or Left / Right Arrows	Adjust horizontal position dynamically on ground
Jump / Midair Jump	Space / W / Up Arrow	Perform ground jump & fluid mid-air double jump
Slide Under Pipes	S / Down Arrow	Slide under overhead clearance pipes & hazards
Slash Action	J / F / Z / Left-Click	Slash attack to break barriers and hit items
Pause Game	P	Toggle pause menu overlay
Quick Restart	R	Instant restart from current level start

4. Educational Mechanics & Boss Battles

RE-RUN seamlessly integrates academic challenges into high-speed runner gameplay:

- **Mario-Style Question Blocks:** Hitting floating Question Blocks triggers interactive pixelated quiz modals testing Computer Science, Data Structures, Algorithms, and Cybersecurity.
- **Departmental Professor Bosses:** At the end of each level (1100m distance), a Departmental Boss (e.g. Prof. Sterling) spawns. The boss launches flying exam paper projectiles while presenting the Final Exam Quiz.
- **Lifelines & Revival:** Players start with 3 Lifelines. Sustaining damage respawns the player slightly ahead on safe track. Falling below 0 lifelines allows instant revival using 100 collected coins.



Figure 2: Final Exam Boss Encounter with Prof. Sterling (Left) & Academic Quiz Modal (Right)

5. Character Roster & Asset Architecture

All character models are extracted exclusively from **KW_School_Character_Set_Free.zip**. The game provides full character selection liberty across **22 Student Characters**:

Category	Count	Featured Student Characters & Majors
Male Students	11 Characters	Leo Tanaka (CS), Kai Sterling (Cybersec), Ren Takahashi (Game Dev), Jin Mori (DevOps), Arata Sato (Robotics), Daiki Kudo (Graphics), Haruto Kuroda (Compilers)...
Female Students	11 Characters	Maya Lin (AI & Data), Chloe Vance (Software Eng), Aoi Sato (UI/UX), Hana Mori (Data Science), Yui Tanaka (Embedded)...
Faculty / Bosses	5 Teachers	Prof. Sterling (Algorithms), Prof. Vance (Cybersecurity), Prof. Kuroda (Systems Architecture)...

6. Installation & Execution Guide

To run the RE-RUN game executable locally:

1. Clone or download the repository: <https://github.com/stewin16/RE-RUN.git>
2. Double-click **PLAY_GAME.bat** in the project root directory.
3. Alternatively, launch via command line: `godot_bin\Godot_v4.3-stable_win64_console.exe --path RE-RUN\RE-RUN`